

**INDIRA GANDHI NATIONAL OPEN
UNIVERSITY
BACHELOR OF COMPUTER APPLICATIONS
Synopsis
AMBULANCE DISPATCH SYSTEM**

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INTRODUCTION

The Ambulance Dispatch System is an innovative application designed to streamline and expedite emergency medical services. With a simple click, witnesses of accidents or individuals in need of urgent medical assistance can request an ambulance through this user-friendly mobile application. The primary objective of this project is to ensure the prompt arrival of the nearest available ambulance to the user's location, optimizing emergency response times and potentially saving lives.

By leveraging location tracking technology, the application identifies the user's precise location and retrieves real-time data from a comprehensive ambulance database. Using a sophisticated dispatch algorithm, the system evaluates the availability and proximity of ambulances, promptly dispatching the nearest one to the user's location.

By combining technology, efficiency, and accessibility, the Ambulance Dispatch System aims to enhance emergency response services, improving overall public safety and facilitating faster medical assistance during emergencies.

OBJECTIVES

- **Improve Emergency Response Time:** The primary goal of this project is to enhance emergency response times by quickly dispatching the nearest available ambulance to the location of the emergency. By reducing the time it takes for medical assistance to reach the scene, the system aims to improve patient outcomes and potentially save lives.
- **Optimize Ambulance Allocation:** The project aims to optimize the allocation of ambulances by efficiently assigning them based on their proximity and availability. By utilizing a dispatch algorithm, the system ensures that ambulances are utilized effectively and dispatched in the most efficient manner.
- **Enhance User Experience:** The project focuses on providing a user-friendly interface that allows witnesses of accidents or individuals in need of medical assistance to request an ambulance with a single click. The aim is to make the process as simple and intuitive as possible, reducing stress during emergencies.
- **Increase Efficiency in Emergency Services:** By streamlining the ambulance dispatch process, the project aims to increase the overall efficiency of emergency services. The system minimizes delays and ensures that the right resources are allocated promptly, improving the overall effectiveness of emergency medical response.
- **Foster Accountability and Feedback:** The project aims to establish a feedback mechanism within the application, allowing users to provide ratings and feedback on the dispatched ambulance services. This fosters accountability, helps improve the quality of service, and provides valuable insights for ongoing system enhancements.

PROJECT CATEGORY

AMBULANCE DISPATCH SYSTEM which is a system is done using android application and web application

This app helps to find near by ambulance for any emergency cases. So it make everything efficient and effective in our life.

HARDWARE AND SOFTWARE REQUIREMENT

This specifies the hardware and the support software required to carry out the development

Hardware Specifications

The selection of hardware is very important in the existence and proper working of any software. Then selection hardware, the size and capacity requirements are also important.

- Processor : Intel Pentium Core i3 and above
- Primary Memory : 4GB RAM and above
- Storage : 320 GB hard disk and above
- Display : VGA Color Monitor
- Key Board : Windows compatible
- Mouse : Windows compatible

Software Specifications

One of the most difficult task is selecting software for the system, once the system requirements is found out then we have to determine whether a particular software package fits for those system requirements. The application requirement:

- Front end : HTML(web)
- Language : Python
- Back end : MySQL
- Operating system : windows 7 or above
- IDE : Pycharm , Android studio

PROPOSED SYSTEM

The system proposed here will help the users book an ambulance easily in an instant. The user will have to select the ambulance. In case of emergency, the user will have to just select the pick-up point & destination and the system will automatically book the nearest ambulance. Once booked the ambulance operator will receive a notification for confirmation of the booking. The Ambulance driver can view the pick-up and drop location on Google Maps. The users will receive the contact details of the driver. The Hospitals can also view the booking history. This is how this Ambulance Booking App will act as a life savior in times of medical emergency.

MODULE

ADMIN

- login
- verify new ambulance
- view verified ambulance/block
- view user list
- view complaints and send reply
- view emergency request
- view rating of ambulance

AMBULANCE/DRIVER

- Register
- Login
- View profile and update
- View emergency request and verify / Track location
- View Verified request and update status
- View previous request
- View rating
- Update location

USER

- Register

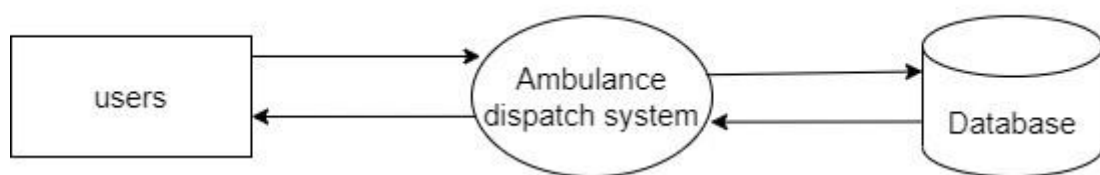
- Login
- View profile and update
- Send emergency request
- View emergency request status
- View previous request
- Send complaint and View reply
- Send rating

DATA MODELING

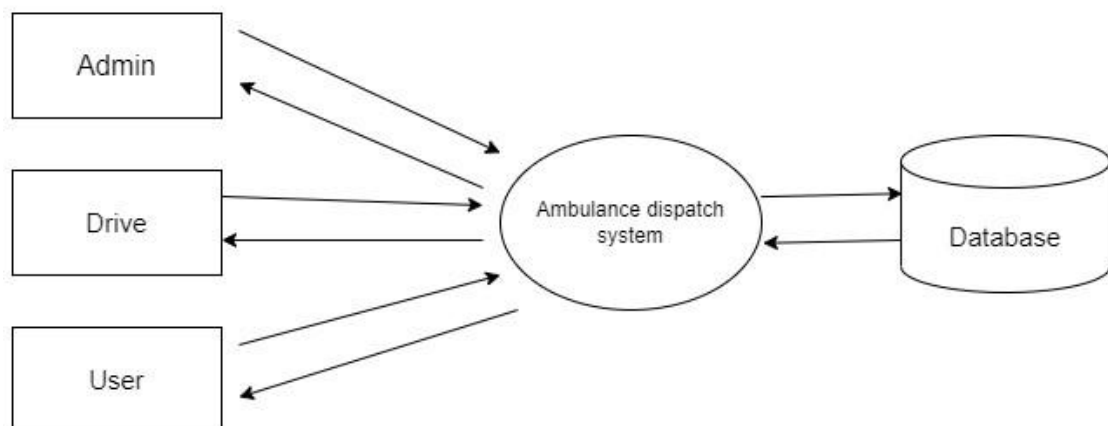
Data modeling defines primary data objects, composition of each data object, and attributes of the object, relationships between each object and other objects and between objects and the process.

DATA FLOW DIAGRAM

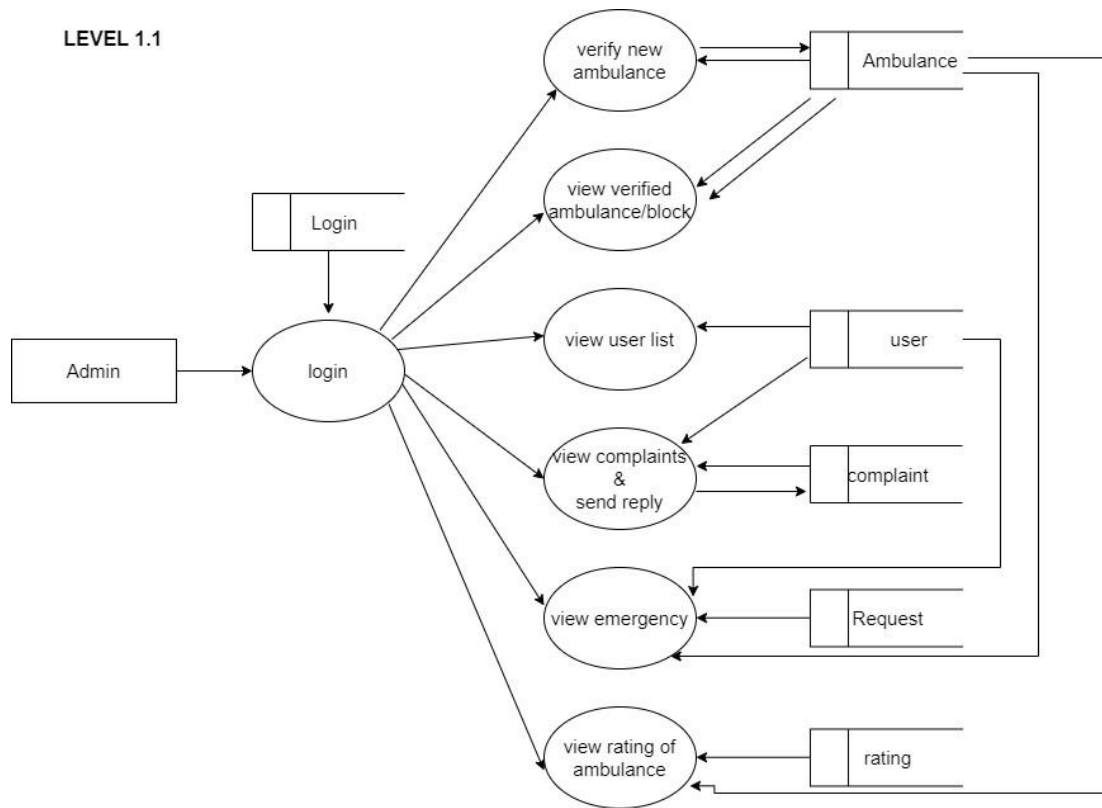
LEVEL 0



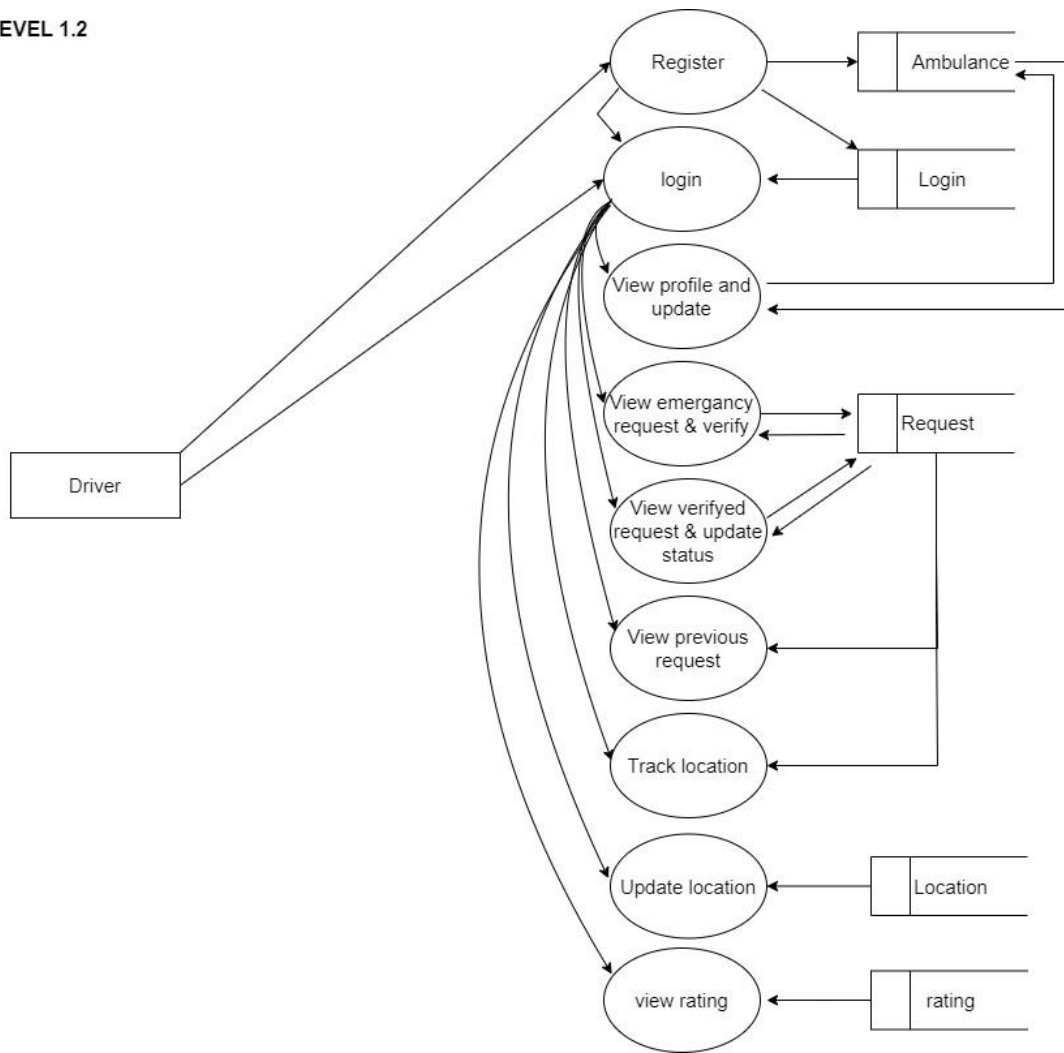
LEVEL 1



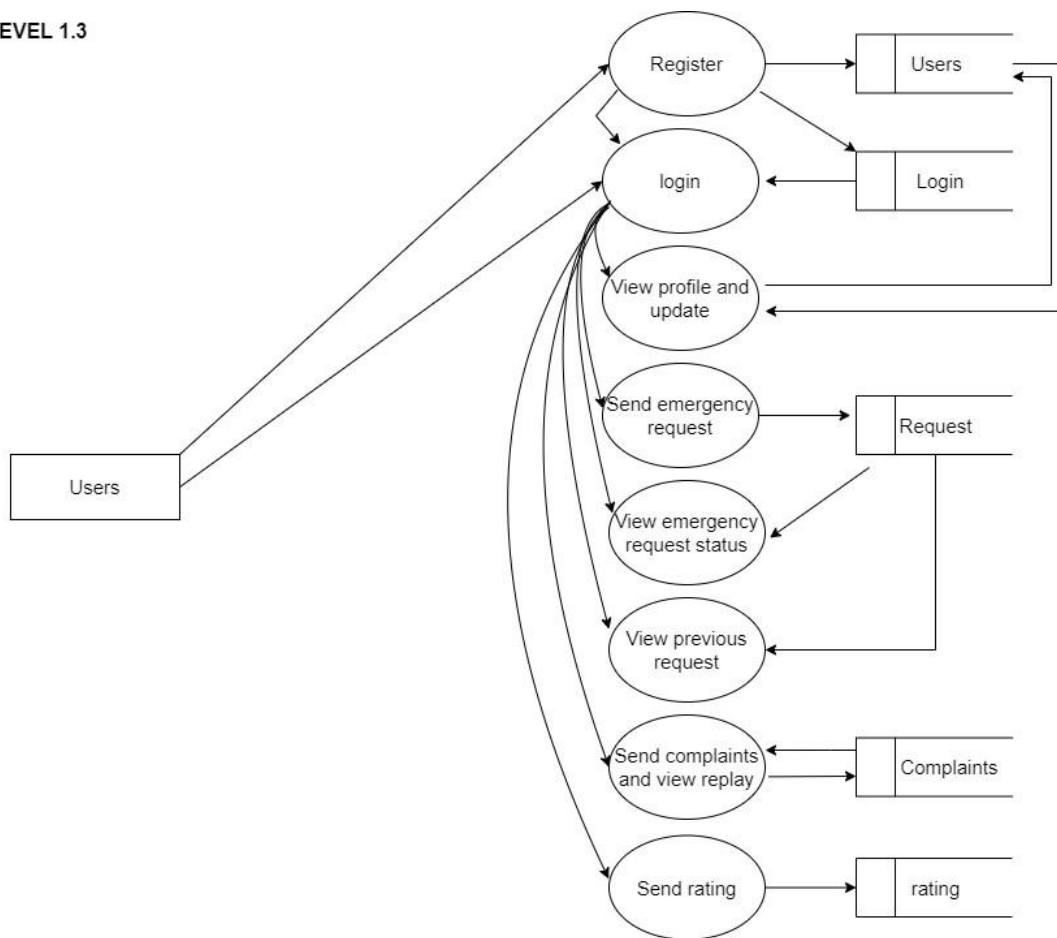
LEVEL 1.1



LEVEL 1.2



LEVEL 1.3



ER DIAGRAM

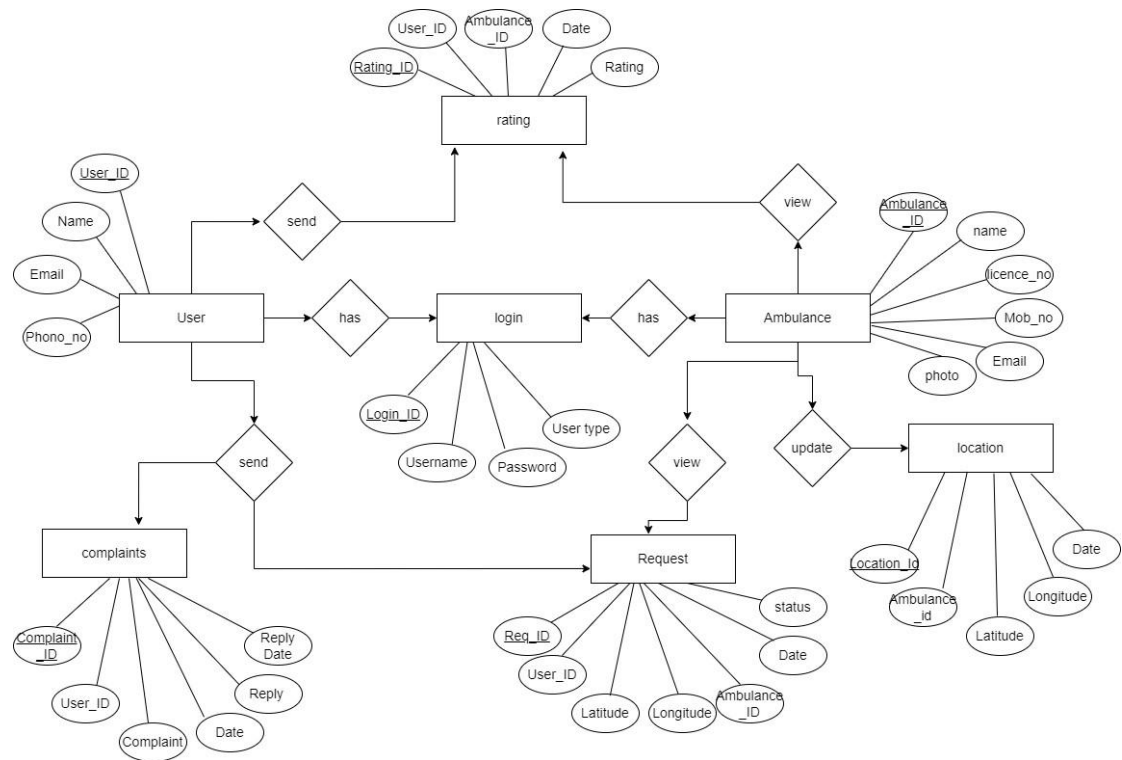


TABLE DESIGN

Table name: LOGIN

FIELD	TYPE	CONSTRAINS	DESCRIPTION
Login_ID	INT(10)	Primary key	Login ID
username	VARCHAR(100)	Not null	Username of admin
Password	VARCHAR(20)	Not null	Password of the admin
User type	VARCHAR(20)	Not null	Type of the user

Table name: AMBULANCE

FIELD	TYPE	CONSTRAINS	DESCRIPTION
Ambulance_ID	INT(10)	Primary key	Ambulance_id
Name	VARCHAR(20)	Not null	Name of the driver
Licence_no	VARCHAR(20)	Not null	Licence number of the driver
Mob_no	BIG INT	Not null	Mobile number
Email	VARCHAR(200)	Not null	email
Photo	VARCHAR(200)	Not null	Photo of the driver

Table name: REQUEST

FIELD	TYPE	CONSTRAINS	DESCRIPTION
Req_ID	INT(10)	Primary key	Request id
User_ID	INT(10)	Foreign key	User_id
Ambulance_ID	INT(10)	Foreign key	Ambulance_id
Date	Date	Not null	Date
Status	VARCHAR(20)	Not null	Current Status of the ambulance
Latitude	INT(10)	Foreign key	Latitude
Longitude	DATE	Not null	Longitude

Table name: LOCATION

FIELD	TYPE	CONSTRAINS	DESCRIPTION
Location_ID	INT(10)	Primary key	Location id
Ambulance_ID	INT(10)	Foreign key	User id
Latitude	INT(10)	Foreign key	Latitude
Longitude	DATE	Not null	Longitude
Status	VARCHAR(20)	Not null	Current Status

Table name:RATING

FIELD	TYPE	CONSTRAINS	DESCRIPTION
Rating_ID	INT(10)	Primary key	Rating ID
User_ID	INT(10)	Foreign key	User ID
Ambulance_ID	INT(10)	Foreign key	Ambulance_id
Rating	VARCHAR(20)	Not null	Rating of ambulance
Date	DATE	Not null	date

Table name: USER

FIELD	TYPE	CONSTRAINS	DESCRIPTION
User_ID	INT(10)	Primary key	User ID
Name	VARCHAR(100)	Not null	Name of the user
Email	VARCHAR(20)	Not null	email
Phone_no	VARCHAR(20)	Not null	Phone number

Table name: COMPLAINT

FIELD	TYPE	CONSTRAINS	DESCRIPTION
Complaint_ID	INT(10)	Primary key	Complaint_ID
User_ID	INT(10)	Foreign key	User_ID
Complaint	VARCHAR(20)	Not null	Complaint by user
Date	DATE	Not null	Date
Reply	VARCHAR(200)	Not null	Reply
Reply date	DATE	Not null	Reply date

TESTING

Software testing is a critical element of software quality assurance and represents the ultimate review of the specification, design and coding. Testing ensure that the system as a whole is bug free. Testing perform a critical role for quality assurance and for ensuring the reliability of the software.

In order to find a highest possible number of errors, tests must be conducted systematically and test cases must be designed using disciplined and techniques. The objective of the software testing is to uncover errors. To fulfill these objectives, a series of testing strategies are used.

They are,

1. Unit Testing
2. Integration Testing
3. System Testing
4. Acceptance Testing
5. Validation Testing

Unit Testing:

Unit testing is carried out to screen wise, each screen being identified as an object. Attention is diverted to individual modules, independently to one another to locate errors in coding and logic.

In unit testing,

Module interface is tested to ensure that information properly flows into and out of the program under test.

Local data structures are examined to ensure that data stored temporarily

maintains its integrity during all steps in algorithm execution.

Boundary condition is tested to ensure that the module operates properly at boundaries established to limit or restricted processing.

All independent paths through the control structures are executed to ensure that all statements in the module have been executed at least once.

Error handling paths are also tested. This test focuses verification effort on the smallest unit of software design, the module. Here the module interface, boundary conditions, and all independent paths were verified by inputting false data. Each single operation is tested individually for its correct functionality.

Integration Testing

Integration testing is a systematic technique for the constructing the program structure while at the same time conducting tests to uncover errors associated with interfacing. Unit tested module were taken and a single program structure was build that has been dictated by the design. Incremental integration has adopted here. The entire software was developed and tested in a small segment, where errors were easy to locate

and rectify. Each database or table manipulation operation was written as a single java class files and separately tested and combined form a single program. After integration, the single program was tested again with numerous test data to check for its functionality.

The integration can be performed in two ways:

Top down integration.

Bottom up integration

Here bottom up integration is performed. This strategy is implemented with the following

steps.

1. Low-level modules are combined to form clusters that perform a specific software sub function.
2. The cluster is tested.
3. Drivers are removed and clusters are combined moving upward in the program structure.

Validation Testing

After performing validation testing, the software was completely assembled as package, interfacing errors have been uncovered and corrected, and a final series of software validation testing began. Here we test the system function in a manner that can be reasonable expected by the customer; the system was tested against the system requirement specification.

Output Testing

After performing validation test, the next phase is output test of the system, since no system could be useful if it is does not produce the desired output in the desired format. By considered the format is considered in to two ways: one is on the screen and the other is on the printed form.

System Testing

System testing mainly concerned with the software-based systems. This actually a series of different tests whose primary purpose is to fully exercise the computer-based systems. Although each test as different purpose all work should be verify that all system elements have been properly integrated and perform allocated functions.

Acceptance Testing

User acceptance test of a system is the key factor for the success of the system. The system under consideration was listed for the user acceptance by keeping constant touch with the perspective user of the system was at the time design, development and making change whenever required. This is done with the regards of the following points:

Input screen design

Output design

Outline message

Menu driven systems

Formats of reports

SCOPE OF THE PROJECT

The scope of the Ambulance Dispatch System project encompasses the development and implementation of a robust and user-friendly application that facilitates efficient ambulance dispatch and emergency response. The system aims to address the following key aspects:

Firstly, the project will focus on creating a user interface that allows witnesses of accidents or individuals in need of immediate medical assistance to request an ambulance quickly and effortlessly. The application will provide a seamless and intuitive user experience, ensuring that users can easily access the emergency services with minimal effort. The interface will be designed to be accessible to users of different age groups and technological backgrounds.

Secondly, the project will involve the integration of location tracking and mapping functionalities. The system will leverage GPS technology to accurately track the user's location in real-time. This information will be utilized to identify the nearest available ambulances and display their locations on a map interface. This feature will enable both users and emergency responders to efficiently navigate to the accident scene, reducing response times and improving overall emergency response effectiveness.

Lastly, the project will encompass the development of an intelligent dispatch algorithm. This algorithm will consider factors such as ambulance availability, proximity to the user's location, and estimated arrival times to determine the most suitable ambulance for dispatch. In cases where the nearest ambulance is occupied, the algorithm will intelligently redirect the request to the next available ambulance, ensuring that no time is wasted during critical situations. The dispatch algorithm will be designed to optimize ambulance allocation and enhance emergency response efficiency.

REPORT GENERATION

The application software is titled “**AMBULANCE DISPATCH SYSTEM**”. The tailormade software which suites the requirements of a calling a near by ambulance, is developed using Python and HTML as front end and MYSQL as the back end.

This application allow users to call near by ambulance by a simple click on the application,So, this application is planned to develop on the requirements of the client

PROJECT SHEDULING

Tasks	Week 1	Week 2	Week3	Week 4
a) Requirement Analysis b) Report Generation Milestone				
a) Input/Output Design b) File Design c) Processing Logic Design d) Design of Control Milestone				
a) Coding Milestone				
a) Module Testing b) System Testing Milestone				

CONCLUSION

At the final stage of the project, we come into a conclusion that the various stages and process of the construction of our proposed system **"AMBULANCE DISPATCH SYSTEM"** has been successfully completed.

As the various aspects falls in place one by one, we can conclude that the final application has been crossed successfully.

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